

LAB 1 — University Management System using DDL Commands

Aim : To design a **University Management System** using DDL commands in MySQL, including CREATE, ALTER, and DROP.

```
CREATE DATABASE UNIVERSITY_DB;      //Create database
```

```
USE UNIVERSITY_DB;      // Select the database
```

1. CREATE TABLE

```
CREATE TABLE STUDENT (  
    Student_ID INT PRIMARY KEY,  
    Student_Name VARCHAR(50),  
    Age INT,  
    Course VARCHAR(50)  
);  
  
DESC STUDENT;      // Check the structure
```

Output:

Output:

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Student_Name	varchar(50)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(50)	YES		NULL	

```
CREATE TABLE DEPARTMENT (  
    Department_ID INT PRIMARY KEY,  
    Department_Name VARCHAR(50),  
    HOD VARCHAR(50)  
);
```

DESC DEPARTMENT;

Output:

Field	Type	Null	Key	Default	Extra
Department_ID	int	NO	PRI	NULL	
Department_Name	varchar(50)	YES		NULL	
HOD	varchar(50)	YES		NULL	

2. ALTER TABLE

A. ADD : Add an email column to the STUDENT table.

ALTER TABLE STUDENT

ADD Email VARCHAR(100);

DESC STUDENT;

Output:

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Student_Name	varchar(50)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(50)	YES		NULL	
Email	varchar(100)	YES		NULL	

B. MODIFY: Change the size of the Course column from VARCHAR(50) to VARCHAR(100).

ALTER TABLE STUDENT

MODIFY Course VARCHAR(100);

DESC STUDENT;

Output:

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Student_Name	varchar(50)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(100)	YES		NULL	
Email	varchar(100)	YES		NULL	

C. RENAME: Rename the column Student_Name to Full_Name.

```
ALTER TABLE STUDENT
```

```
RENAME COLUMN Student_Name TO Full_Name;
```

```
DESC STUDENT;
```

Output:

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Full_Name	varchar(50)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(100)	YES		NULL	
Email	varchar(100)	YES		NULL	

D. DROP: Remove the Email column.

```
ALTER TABLE STUDENT
```

```
DROP COLUMN Email;
```

```
DESC STUDENT;
```

Output:

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Full_Name	varchar(50)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(100)	YES		NULL	

3. DROP TABLE: Delete the DEPARTMENT table.

DROP TABLE DEPARTMENT;

SHOW TABLES;

Output:

Tables_in_sandbox_db
student

Whole Code:

OneCompiler  AI MySQL Run

```

1  -- Create
2  CREATE TABLE STUDENT (
3      Student_ID INT PRIMARY KEY,
4      Student_Name VARCHAR(50),
5      Age INT,
6      Course VARCHAR(50)
7  );
8  DESC STUDENT;
9
10 CREATE TABLE DEPARTMENT (
11     Department_ID INT PRIMARY KEY,
12     Department_Name VARCHAR(50),
13     HOD VARCHAR(50)
14 );
15 DESC DEPARTMENT;
16
17 -- Alter
18 ALTER TABLE STUDENT
19 ADD Email VARCHAR(100);
20 DESC STUDENT;
21
22 ALTER TABLE STUDENT
23 MODIFY Course VARCHAR(100);
24 DESC STUDENT;
25
26 ALTER TABLE STUDENT
27 RENAME COLUMN Student_Name TO Full_Name;
28 DESC STUDENT;
29
30 ALTER TABLE STUDENT
31 DROP COLUMN Email;
32 DESC STUDENT;
33
34 -- drop
35 DROP TABLE DEPARTMENT;
36 SHOW TABLES;

```

Result

Thus, a **University Management System** was designed successfully using DDL commands such as CREATE, ALTER (ADD, MODIFY, RENAME, DROP) and DROP TABLE in MySQL.